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UNIVERSITY OF COLOMBO, SRI LANKA

UCSC UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING
07 APR 2025
EXAMINATIONS & REGISTRATION

# BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

**First Year Examination - Semester 2 - UCSC AY21 [held in March/April 2025]**

## *IS 1208 - Systems Analysis and Design*

**Two (2) HOURS**

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**Answer All Questions**

**Number of Pages: 12** **Number of Questions: 4**

To be completed by the candidate

Index Number: | | | | | | | | |

**Important Instructions to candidates:**

I. Students should answer in the medium of English language only using the space provided in this question paper.

II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.

III. Write your index number CLEARLY on each and every page of this Question paper.

IV. This paper consists of 4 questions in 12 pages (including the Cover Page).

V. Answer ALL questions.

VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.

VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

**For Examiner's use only**

| Question No | Marks |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| Total | |

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Q1. Each of the following MCQ (Multiple Choice Questions) will have four (4) choices with **one or more correct answers**. Each question will carry equal marks. There will be a penalty for incorrect responses to discourage guessing. Circle or underline the correct answer(s).

(1) Which of the statements given below is/are correct regarding IT professionals?

(a) A Network Administrator manages and maintains computer networks, ensuring network security and performance.
(b) A Business Analyst acts as a bridge between the technical team and business stakeholders, understanding business needs and translating them into technical requirements for IT solutions.
(c) A Data Scientist analyzes and interprets complex data sets, uses data visualization tools and develops machine learning models.
(d) A Systems Analyst designs, develops, and tests software applications.

(2) Some questions related to modeling with possible answers are given below.

(i) Q. What is the primary purpose of a context-level DFD?
A. To show how the system interacts with external entities.

(ii) Q. What is a *Process* in a DFD in system analysis and design?
A. A *Process* represents a specific function or operation within the system that transforms incoming data into outgoing data, and it essentially illustrates how data is processed or manipulated within the system.

(iii) Q. What is an External Entity in a DFD?
A. It is a place where data is stored within the system, such as databases or files.

Which of the above pairs is/are correct?
(a) Only (i) (b) Only (i) and (ii) (c) Only (ii) and (iii) (d) All

(3) Video conferencing has become an integral tool for communication, especially with the rise of remote work and virtual meetings.

Which of the following is/are considered as **disadvantage(s)** of using video conference facility?

(a) Technical issues like poor connectivity, lag, and glitches can disrupt meetings, causing misunderstandings and reduced productivity.
(b) Security risks include hacking and data breaches, compromising sensitive information.
(c) Integration with project management and productivity tools may be limited, affecting workflow efficiency.
(d) Lacks in-person interaction, making non-verbal signals harder to interpret, affecting communication and relationships.

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(4) Which of the following statements is/are correct regarding Software Process Models?

(i) The V-Model is suitable for situations where the requirements and understanding of the software's functionality are well-defined from the beginning.
(ii) The primary focus of the V-Model in software development is verification and validation
(iii) The V-Model does not inherently support iterative development or the concept of refining the product through multiple cycles.

Which of the above statements is/are correct?

(a) Only (i) (b) Only (i) and (ii) (c) Only (i) and (iii) (d) All

(5) Consider the following Requirements.

(i) The system should respond to user requests within 2 seconds.
(ii) The system must allow users to reset their passwords via email.
(iii) The system should allow users to view their order history and details.
(iv) The system should display suggestions as users type their queries in the search bar.
(v) The system should support scaling up to 10,000 concurrent users without performance degradation.

Identify which of the above are **non-functional requirements (NFR)** and which are **functional requirements (FR)**.

(a) NFR-(i), (ii),(iii) : FR –(iv),(v)
(b) NFR-(ii),(iii) : FR –(i), (iv),(v)
(c) NFR-(ii),(iii), (iv) : FR –(i),(v)
(d) NFR (i),(v) : FR –(ii),(iii),(iv)

(6) Which of the following is/are correct regarding the User Stories?

(a) A User Story typically follows the format:
*"As a [user role], I want to [action], so that [benefit]."*
(b) Epics are large User Stories that are broken down into smaller user stories for easier implementation.
(c) When a User Story is too complex to complete in one sprint it should be split into smaller stories.
(d) Developer A completed 10 story points, while Developer B completed 5, making Developer A appear more productive.

(7) Which of the following statements regarding Systems Analysis and Systems Design is/are correct?

(a) Systems Analysis involves defining the architecture, components, modules, and data structure of the system.
(b) Systems Design is focused on identifying problems, gathering requirements, and analyzing system needs.
(c) Key activities of Systems Design include designing databases and interfaces, defining system components and modules, and selecting hardware/software requirements.
(d) Key activities of Systems Analysis involve gathering and analyzing requirements, identifying user needs, understanding business processes, and evaluating existing systems.

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(8) Which of the following is/are true regarding *Conceptual and Structural models*?
(a) The main purpose of conceptual modeling is to describe the high-level structure of a system without technical details.
(b) Conceptual models include technical or system-specific details.
(c) The conceptual model provides a schema for how the data will be stored physically within a database.
(d) Class diagrams represent the structure of a system and are a key part of structural modeling.

(9) Which of the following is/are correct regarding categories of feasibility tests?
(i) Common types of feasibility studies include technical, economic, operational, legal, and scheduling feasibility.
(ii) Technical feasibility focused on whether the project can be developed using existing technology and expertise
(iii) Operational feasibility evaluates how well the new system meets user needs and whether it will be accepted by the users.
Which of the above statement(s) is /are true?
(a) Only (ii) (b) Only (ii) and (iii) (c) Only (iii) (d) All

(10) Which of the following statements is/are correct regarding CASE tools?
(a) CASE tools are a set of software application programs used to automate SDLC activities.
(b) Classic CASE tools require a central repository, which serves as a central storage location for product specifications, requirement documents, related reports, diagrams, etc.
(c) Real CASE tools can be broadly divided into categories based on their use at specific stages of the SDLC: Upper CASE tools, Lower CASE tools, and Integrated CASE tools
(d) Integrated Case Tools are helpful in all the stages of SDLC, from requirement gathering to testing and documentation.

(11) Consider the following statements with respect to the statement: "Factoring, partitions and collaborations, and layers can be used to evolve problem domain-oriented analysis models into optimal solution domain-oriented design models."
(i) In object-oriented (OO) design, *factoring* refers to the process of breaking down complex systems or designs into smaller, more manageable, and more reusable parts.
(ii) An easy approach to model *partitions* and *collaborations* is the use of packages and package diagrams.
(iii) *layers* refer to the logical separation of components or modules based on their functionality, responsibilities, or level of abstraction.
Which of the above statements is/are correct?
(a) Only (i) (b) Only (ii) (c) Only (i) and (ii) (d) All

(12) Some questions related to UML deployment diagrams with possible answers are given below.
(i) Q. Describe the usage of deployment diagrams.
A. To visualize the system's physical architecture, showing how components map to hardware nodes, and to understand software distribution and communication across machines.

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(ii) Q. How do you represent a *node* in a deployment diagram?

A. A *node* is represented as a 3D rectangular box and it symbolizes physical or virtual hardware that hosts software components in the system.

(iii) Q. What kind of information can appear on a node?

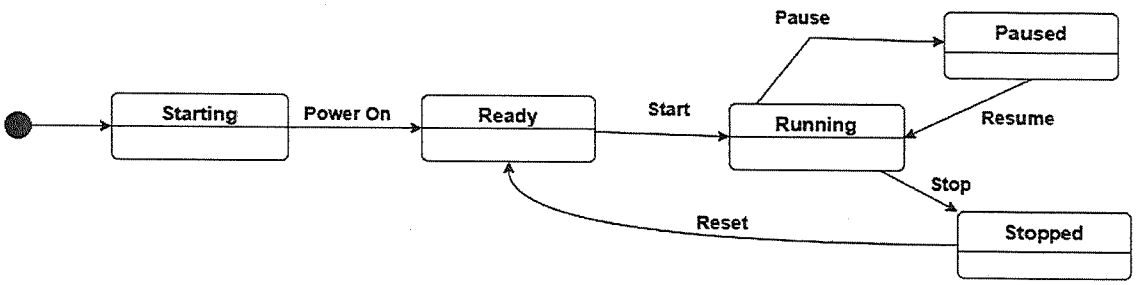
A. A node represents physical or virtual hardware and contains information about its characteristics, hosted software, and interactions with other nodes.

Which of the above question-answer pairs is/are correct?

(a) Only (i) (b) Only (ii) (c) Only (i) and (ii) (d) All

(25 Marks)

Q2. (a) Consider the following State Diagram and the given statements.




(i) The system can have a transition directly from "Stopped" to "Running".

(ii) The "Paused" state can only be reached from the "Ready" state.

(iii) The system cannot have a transition from "Ready" to "Stopped" without passing through the "Running" state.

(iv) A Paused system can resume running, while a Stopped system requires a Reset to return to "Ready."

(v) "Power On", "Start", "Stop", "Reset", and "Resume" are events that cause state changes in the system.

Which of the above statements is/are correct according to the given diagram? Write the corresponding number of the correct statement(s).

(10 marks)

(b) Draw a UML class diagram to illustrate the following relationships clearly representing the multiplicity.

i) Model the relationship between a 'Person' and a 'Company' where a person can be employed by multiple companies and each company can employ multiple people.

ii) Since employment details differ for each job, include an **association class** to store attributes such as salary and startDate for each employment instance.

(10 Marks)

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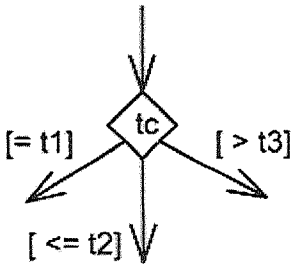
(c) State whether the following statement is correct.
"A guard expression (tc) in an Activity diagram cannot be represented as given below. It can have only two possible alternatives."
Justify your answer if it's incorrect.




(05 Marks)

**Q3.** (a) Write down **two (2)** advantages and **two (2)** disadvantages of Model Driven Engineering.
(04 Marks)

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(b) **Correctly match the models in Column X with their corresponding descriptions in Column Y.** *(3 Marks)*

| X | Y |
|---|---|
| A. Computation Independent Model (CIM) | (i) UML Class Diagrams, State Diagrams, Sequence Diagrams without platform details |
| B. Platform Independent Model (PIM) | (ii) UML Class Diagrams with Java-specific attributes and SQL database schemas |
| C. Platform Specific Models (PSM) | (iii) Captures domain knowledge, business processes, and stakeholder needs |

(c) What is Hybrid Inheritance? Illustrate using an example. (05 Marks)

(d) Create a *State* diagram illustrating the cardiovascular risk calculation process within the web application. The process consists of several states: Input Received, Data Preprocessing, Model Execution, Risk Level Assigned, and Results Output.
The transitions between these states occur as follows:
When the API receives user input, the system moves from Input Received to Data Preprocessing. During Data Preprocessing, input data is normalized, leading to a transition to Model Execution. In this state, a trained model calculates the probability of cardiovascular disease (CVD), which then transitions to Risk Level Assigned. Here, the probability is mapped to a specific risk category before proceeding to Results Output, where the final results are formatted and sent to the user. The activities performed in each state include normalizing input data and ensuring consistency in Data Preprocessing, loading the trained model and predicting CVD probability in Model Execution, mapping the probability to WHO risk categories in Risk Level Assigned, and formatting and delivering results in Results Output.

(09 Marks)

(e) What is a Composite Structure Diagram? Draw a composite structure diagram that represents a 'Person' composed of a 'Mind' and a 'Body'.

(04 Marks)

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Q4. Pasan Motors, a company specializing in importing and selling motor vehicles, requires an Information System to streamline its operations. Currently, the company deals with brand-new and re-conditioned Motor Cars and SUVs, with plans to expand its sales to passenger vans and motor coaches in the future.

*Vehicle Information Management:* Each vehicle in the system must be recorded with key details, including Chassis Number, Engine Number, Make (e.g., Toyota, Nissan), Model (e.g., Axio, Allion), Year of Manufacture, Country of Manufacture, Cylinder Capacity (e.g., 1.5cc, 1.8cc), Vehicle Type (e.g., Motor Car, SUV), Fuel Type (Petrol, Diesel), Mileage, Purchased Price, and Selling Price.
Additionally, vehicles may include various optional features such as power mirrors, power shutters, retractable mirrors, cruise control, multifunction keys, and push-start systems. A single vehicle may have multiple optional features.

*Customer and Payment Management*: Customer information, including NIC/Passport Number, Name, Address, and Contact Number, must be stored in the system. Customers can purchase vehicles using different payment methods, and a payment clerk enters the payment details into the system.

*Vehicle Exchange Program*: Pasan Motors allows customers to upgrade their vehicles by exchanging their old ones. The system records the Exchange Value, which is deducted from the price of the new vehicle. Details regarding the exchange transaction are maintained.

*Warranty and Maintenance Management:* Vehicles sold by Pasan Motors come with a warranty period:
Brand-New Vehicles: 50,000 km warranty
Re-conditioned Vehicles: 25,000 km warranty
Customers can claim free repairs if faults occur within the warranty period. The system maintains records of such repairs, which customers can access upon logging in.

*Reporting and Administration*: The Manager of Pasan Motors should be able to generate various reports related to purchases, sales, customer details, vehicle details, purchase requests, and maintenance requests.

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A Registration Clerk is responsible for handling customer registrations, vehicle records, purchase requests, and maintenance requests in the system.

This Information System will enhance Pasan Motors' efficiency by managing vehicle inventory, customer interactions, sales transactions, and post-sales services effectively.

(a) Customer is a System Actor for the above system. Identify **three (3)** responsibilities or use cases for "Customer".

(03 Marks)

(b) Identify **three (3)** more system actors for the above system. Write down **two (2)** responsibilities for **three (3)** actors identified. Draw a Use Case diagram segment to illustrate these responsibilities.

(06 Marks)

(c) 'Vehicle' is one of the identified classes for the above system. Identify **six (6)** methods that could be included in the 'Vehicle' class.

(03 Marks)

(d) Customer, Sales Transactions, Repair, Vehicle, Exchange, and Payment are some of the classes identified in designing the system. Identify the relationships between these classes and draw a UML class diagram by indicating the multiplicities. (10 Marks)

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(e) List down the objects that need to interact during the 'Vehicle purchase' use case?

(03 Marks)

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